

Technical Documentation

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Project Title: Online Retail Sales & Returns Analysis (2009–2011)

Tools Used: Python (pandas, NumPy, Matplotlib, Seaborn), Excel, Tableau Public

1. Project Objective

To explore, clean, and analyze an online retail dataset to uncover revenue trends, customer behavior patterns, return activity, and country-level insights.

- Revenue trends
- Best-performing products and customers
- Country-wise revenue distribution
- Return behavior patterns
- Top returning customers and return rates

2. Dataset Description

- Source: Online Retail transactional dataset (Excel format)
- Period Covered: December 2009 – December 2011
- Features: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country
- Dataset Size: Approximately 500,000 transactions

3. Tools & Libraries

Python

- pandas and NumPy for data cleaning and aggregation
- Matplotlib and Seaborn for exploratory visualizations

Excel

- Prepared clean summary tables for Tableau

Tableau Public

- Built interactive dashboards and Tableau Story

4. Step-by-Step Process

Data Loading & Merging

- Loaded and merged Excel transaction sheets
- Handled inconsistent date formats using `pd.to_datetime(errors='coerce')`

Return Identification: Negative Transactions

Transactions with negative quantity values were treated as product returns.

- Calculated top returned products
- Calculated returns by country
- Calculated monthly return volumes
- Calculated top returning customers

Important Finding: Several values inside the Description field represented operational return reasons rather than valid product names.

- Examples included: 'given away', 'missing', 'check', 'unsaleable, destroyed', and '?'
- These values were retained and documented to avoid misleading interpretation
- Potential use case: operational-quality or anomaly analysis

Data Cleaning

- Removed missing Customer IDs
- Removed invalid negative or zero prices
- Created TotalPrice column ($\text{Quantity} \times \text{UnitPrice}$)

Core Exploratory Data Analysis

- Calculated monthly revenue
- Calculated top products by quantity sold
- Calculated top customers by spending
- Calculated revenue by country
- Applied Euro currency formatting for business readability and executive reporting consistency

Advanced Return Metrics

- Calculated return rate by units
- Calculated return rate by value
- Created ReturnRateUnits customer-level metric

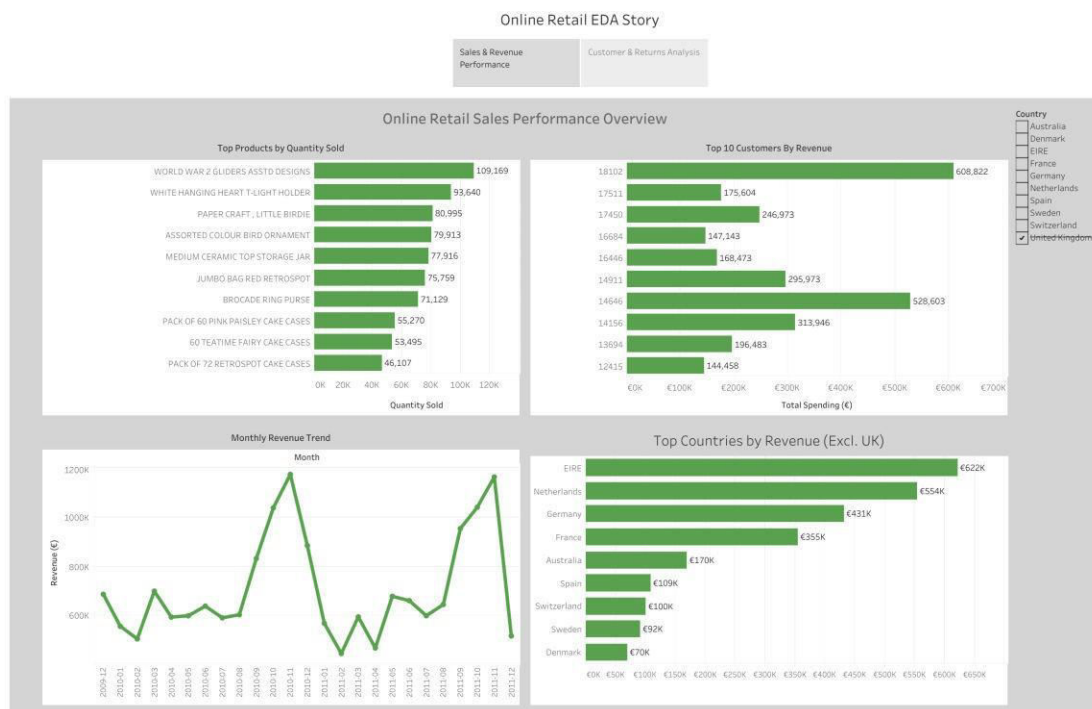
5. Tableau Story Dashboard

The Tableau visualization layer was redesigned into an interactive Tableau Story to improve storytelling, readability, and dashboard navigation.

Sales & Revenue Performance

- Top products by quantity sold
- Top customers by revenue
- Monthly revenue trend
- Top countries by revenue excluding the United Kingdom
- Interactive country filter

The United Kingdom was excluded from one country comparison chart because it dominated revenue and reduced visibility of secondary markets.



Customer & Returns Analysis

- Top returning customers
- Most returned products / return reasons
- Top returning customer KPI
- Total units returned KPI
- Monthly return trend
- Customers with highest return rates
- Interactive month filter

The dashboard clearly distinguishes between return products and operational return reasons.

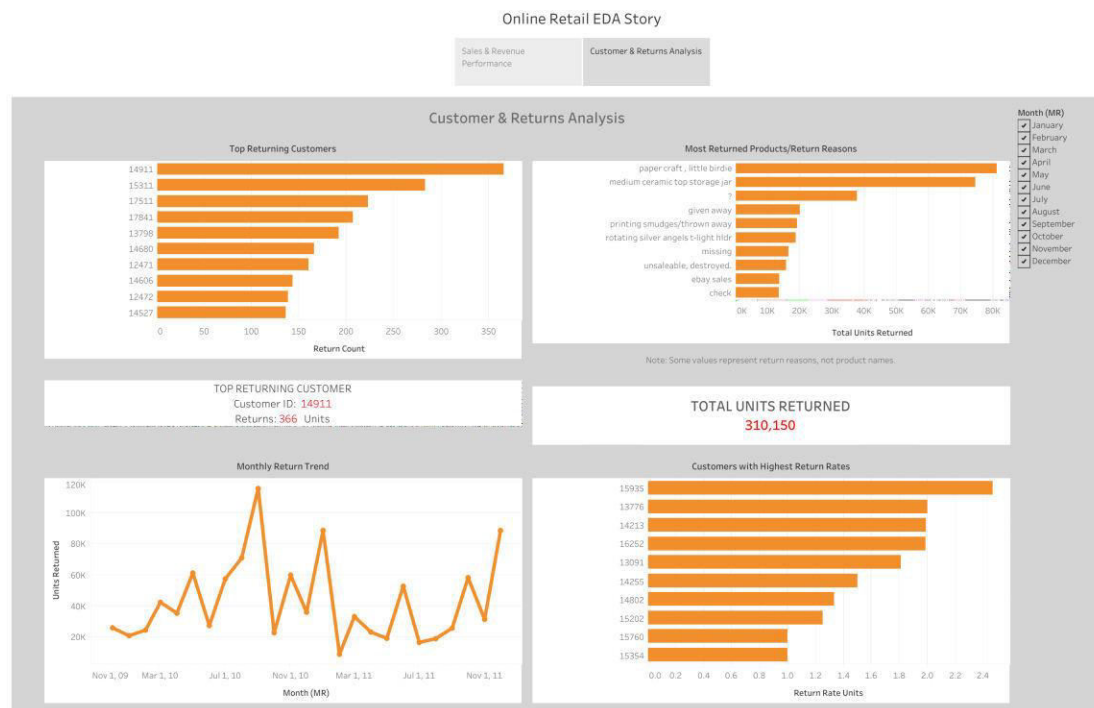


Tableau Story Improvements

- Replaced button-based navigation with Tableau Story navigation
- Added focused dashboard interactivity
- Improved spacing, KPI hierarchy, and visual consistency
- Used green theme for sales and orange theme for returns
- Reduced clutter and improved executive readability
- Exported dashboard previews as PNG files
- Exported full Tableau Story as PDF

6. Challenges Encountered

- Data quality issues including missing values and unclear descriptions
- Dashboard layout issues such as clipping and scrolling
- Tableau field-type mismatches
- Filter design balancing interactivity and readability
- United Kingdom revenue dominance in country analysis

7. Final Deliverables

- Cleaned and transformed Online Retail dataset
- Jupyter Notebook for EDA and preprocessing
- Interactive Tableau Public Story
- Sales & Revenue Performance dashboard

- Customer & Returns Analysis dashboard
- Dashboard PNG previews
- Tableau Story PDF export
- Technical documentation
- Portfolio presentation in PPTX and PDF format
- Mobile Dashboard Layouts
- GitHub repository with reproducible structure

8. Project Summary: Top 3 Insights

1. Revenue Concentration in the United Kingdom

The United Kingdom generated over €14.7M in revenue, making it the dominant market in the dataset.

2. Customer Revenue Concentration

A small group of customers generated a disproportionately large share of revenue. Customer ID 18102 alone generated more than €608K in spending.

3. Return Volume vs. Return Rate

Customers with the highest return volume differed from customers with the highest return rate, demonstrating that return behavior requires multiple evaluation metrics.

9. Bonus Observations

- 'PAPER CRAFT, LITTLE BIRDIE' and 'MEDIUM CERAMIC TOP STORAGE JAR' were both bestselling and highly returned products
- Return spikes in late 2010 and 2011 may indicate seasonal or fulfillment-related effects
- Several unclear return labels indicated operational data-quality issues

10. Lessons Learned

Data Quality Can Mask Hidden Patterns

Several entries in the Description field represented return reasons instead of valid products. Categorical fields should always be validated before visualization.

Tableau Field Types Matter

Currency formatting and incorrect field types can prevent correct dashboard generation.

Clear Filtering Improves Storytelling

Focused filters improved dashboard readability without introducing clutter.

Return Behavior is Multi-Dimensional

Return volume and return rate should both be analyzed to understand customer behavior.

Documentation Improves Reproducibility

Maintaining documentation throughout the project improved clarity and reproducibility.

Dashboard Storytelling Improves Communication

The Tableau Story structure improved business readability and stakeholder navigation.

11. Supporting Files

- dashboards/online_retail_eda_story.twbx
- docs/tableau_exports/tableau_story_export.pdf
- images/Sales_&_Revenue_Performance.png
- images/Customer_&_Returns_Analysis.png